NAME: _____ CLASS: _____

MZUZU DIOCESE

2020 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

PHYSICS

PAPER I
(100 marks)

Theory

Subject Number: M162/I

Thursday, 26th March

Time Allowed: 2hrs
07:30 – 09:30 am

Instructions:

1. **This paper contains 14 pages. Please check.**
2. Fill in your **Name** at the top of each page.
3. Answer **all** the **thirteen** questions in the spaces provided.
4. The maximum number of marks for each answer is indicated against each question.
5. In the table provided on this page, **tick** against the question number you have answered.
6. Hand in your paper to the invigilator when time is called to stop writing.

Question Number	Tick if Answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

Turn over

Name_____ Class_____

Section A

This section contains 70 marks. Answer all questions

1. a) Define angular velocity

_____

_____

(1 mark)

b) A long playing record of diameter 270 mm rotates at $3\frac{1}{3}$ revolutions per minute. Calculate;

i) its angular velocity in radians per second

(3 marks)

ii) the linear velocity of a point along its rim

(3 marks)

2. a) i) State the kinetic theory of matter

_____

_____

(1 mark)

Name_____ Class_____

ii) In terms of Kinetic theory of matter, explain why ice melts when put in the sun

_____

_____ (2 marks)

b) A gas occupies volume of $200cm^3$ at 273k. Calculate the temperature of the gas in Kelvin if its volume increases to $300cm^3$ at constant pressure.

(3 marks)

c) State one factors that affect gas pressure

_____

_____

(1marks)

d) Define Absolute zero

_____

_____

(1 mark)

3. **Figure 1** below is a diagram showing a wave.

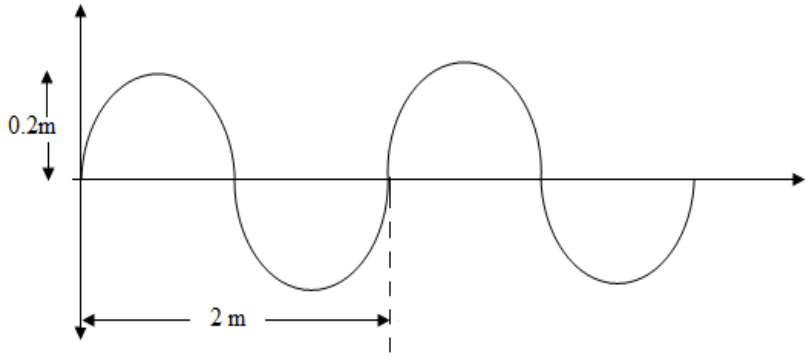

Figure 1

Name_____ Class_____

a) What type of a wave is shown in the diagram?

_____

(1 mark)

b) What is the amplitude of the wave?

_____

(1 mark)

c) Calculate the velocity of the wave if its frequency is 4Hz.

(2 marks)

d) Define the following terms

i) Longitudinal waves

_____

_____

_____(1 mark)

ii) Diffraction of waves

_____

_____

_____(1 mark)

4. a) i) Give two factors that affect the speed of sound in air

_____

_____(2 marks)

ii) Tanangachi standing between two cliffs claps his hands and receives two echoes after 3 minutes and 3.5 minutes consecutively. Determine the distance between the cliffs. Average speed of sound in air = 320 m/s.

(3 marks)

Name_____ Class_____

5. **Figure 2** is a diagram of a transformer connected to an alternating current (AC) power supply.

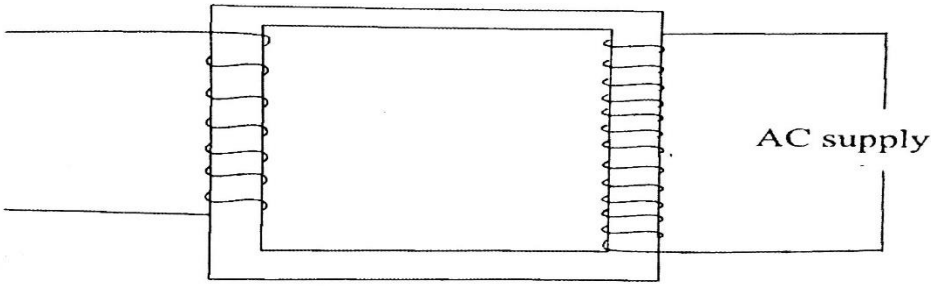

Figure 2

a) What type of transformer is shown in the figure?

_____ (1 mark)

b) If the in-put voltage is 240V, and the number of turns in the secondary coil is 400 and in the primary coil is 800, calculate the output voltage.

(3 marks)

c) State one advantage of alternating current,

_____

_____

(1 marks)

d) Why are "diodes", sometimes referred to as "rectifiers"?

_____

_____

_____ (1 mark)

Name_____ Class_____

e) Draw a circuit diagram consisting of a cell, a bulb and a diode such that the diode is forward biased.

(2 marks)

6. a) You can produce accurate results of an experiment by reducing errors. Give two ways of reducing errors.

_____

_____

(2 marks)

b) Briefly explain systematic errors

_____

_____

_____

(2 marks)

c) **Figure 3** shows a laboratory safety symbol. Give its name and its interpretation

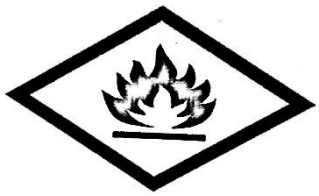

Figure 3

_____

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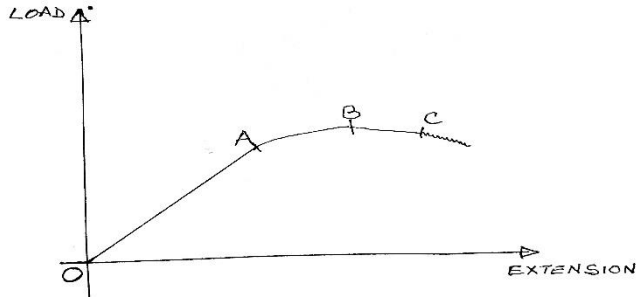
_____

(2 marks)

Name_____ Class_____

7. The graph shows the load plotted against the extension of the wire. Use it to answer the question that follows.



a) Which point is
   i) The yield point _____ (1 mark)
   ii) The elastic limit _____ (1 mark)
   iii) Breaking point _____ (1 mark)

b) What is the force constant of a string which stretched 4cm by a mass of 200g?

(2 marks)

8. **Figure 4** is a diagram of a pulley system being used to lift a mass of 120 kg vertically through a height of 30 m.

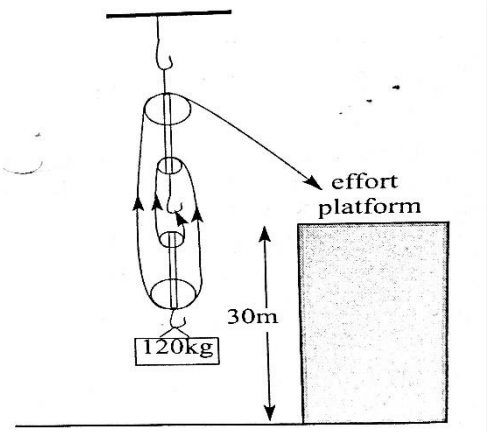


Figure 4

Name_____ Class_____

a) Calculate the work done in lifting the mass to a height of 30 m.

(2 marks)

b) How much energy is supplied by the effort in raising the mass through the height of 30 m?

(1 mark)

c) If it takes 0.5 minutes to complete the work using the pulley system, work out the power of this machine.

(3 marks)

d) i) What do you understand by the term "efficiency"?

_____

_____

(1 mark)

ii) Give two reasons why in practice the machine in figure 1 is not 100% efficient.

_____

_____

_____

(2 marks)

9. Uranium $^{238}_{92}U$, decays in a series as follows

Stage 1: $^{238}_{92}\mathbf{U} \longrightarrow {}^{234}_{90}\mathbf{Th}$

Stage 2: $^{234}_{90}\mathbf{Th} \longrightarrow {}^{234}_{91}\mathbf{Pa}$

Name_____ Class_____

a) What particles are emitted at each stage?

Stage 1 _____

(1 mark)

Stage 2 _____

(1 mark)

b) Apart from the particles mentioned in **5(a) above,** what else is emitted at each stage?

_____

(1 mark)

c) A scintillation counter was used to detect radioactivity of a radioactive substance. The sample recorded 480 scintillations per second. After 8 days the same sample recorded 30 scintillations per second. Work out half life of the sample.

(3 marks)

d) State any two safety precautions that must be followed when handling radioactive substances.

_____

_____

_____

(2 marks)

Name_____ Class_____

10. a) State the law of floatation

_____

_____

_____

(1 mark)

b) The mass of a metal bar is 0.5kg in air and its mass is 0.3kg in water.

Calculate :

i) Apparent loss in weight of the metal bar

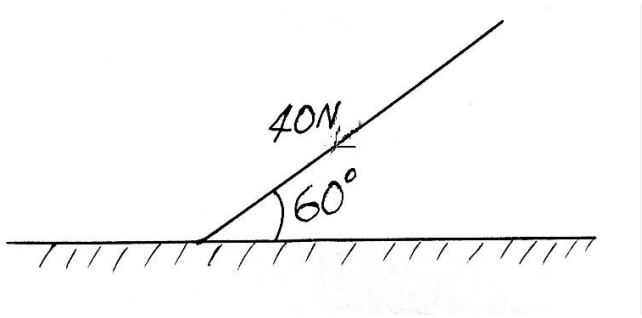
(2 marks)

ii) The upthrust acting on the metal bar

(2 marks)

Name_____ Class_____

c) Figure 6 is a diagram that shows a lawn mower being pushed with a force of 40N acting at an angle of 60° with the ground. Find the horizontal force that is pushing the lawn mower along the ground.



(3 marks)

SECTION B

(30 marks)

Answer all questions in this section.

11.a) Describe an experiment that could be done to determine the effect of impurities on melting and boiling point of substances. In your description include the materials to be used.

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(5 marks)

b) With the aid of a diagram, explain sea breeze.

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(5 marks)

12.a) With an aid of a diagram, describe how you would demonstrate that a material medium is necessary for the transmission of sound.

Name_____ Class_____

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(6 marks)

b) Explain how raising the temperature of semiconductors affects their electrical conductivity.

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(4 marks)

13. a) Briefly explain any one method how you would magnetize an iron bar

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Name_____ Class_____

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(4 marks)

b) Describe how the thickness of a sheet of plastic could be controlled using radiation during manufacture.

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(5 marks)

END OF QUESTION PAPER

NB: This papers contains 14 pages.